

Numerical Analysis chapter 2 theoretical homework

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I

Question1

Solution: The linear interpolation of $f(x)$ at x_0 is given by

$$p_1(f; x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0).$$

As $f(x) = \frac{1}{x}, x_0 = 1, x_1 = 2$ we have:

$$f(x_0) = f(1) = 1,$$

$$f(x_1) = f(2) = \frac{1}{2}.$$

So the linear interpolation of $f(x)$ at $x_0 = 1$ is:

$$p_1(f; x) = 1 + \frac{\frac{1}{2} - 1}{2 - 1}(x - 1) = 1 - \frac{1}{2}(x - 1) = \frac{3 - x}{2}.$$

The error is

$$f(x) - p_1(f; x) = \frac{f''(\xi(x))}{2}(x - 1)(x - 2).$$

As $f(x) = \frac{1}{x}, f''(x) = 2x^{-3}$,

$$\frac{1}{x} - \frac{3 - x}{2} = \frac{\frac{2}{\xi(x)^3}}{2}(x - 1)(x - 2).$$

$$\xi(x) = \sqrt[3]{2x}.$$

□ [1]

Question2

Solution:

$$\max \xi(x) = \xi(2) = \sqrt[3]{4}, \quad \min \xi(x) = \xi(1) = \sqrt[3]{2}, \quad \max f''(\xi(x)) = 1$$

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II

First, we use Lagrange interpolation to construct a polynomial $L(x)$ such that $L(x_i) = f_i$ for $i = 0, 1, \dots, n$. The Lagrange interpolation polynomial can be expressed as:

$$L(x) = \sum_{i=0}^n f_i \ell_i(x)$$

where $\ell_i(x)$ are the Lagrange basis functions:

$$\ell_i(x) = \prod_{\substack{0 \leq j \leq n \\ j \neq i}} \frac{x - x_j}{x_i - x_j}$$

If $p(x) = L(x)$, and $\forall x, p(x) > 0$, then $p(x)$ is what we want. If not, we could add another polynomial $q(x) = A \cdot \prod_{i=1}^n (x - x_i)^2, A > 0$. By adjusting the constant A , it is possible to ensure that $q(x) + L(x) > 0$ for all x , because $q(x)$ is always greater than 0. And this function does not exceed the $2n$ -th order. $\square$

III

Question1

Base Case (n=1):

$$f[t, t+1] = \frac{f(t+1) - f(t)}{(t+1) - t} = \frac{e^{t+1} - e^t}{1} = e^t(e-1).$$

This matches $\frac{(e-1)^1}{1!} e^t$.

Inductive Step: Assume the statement holds for some $k \geq 1$, i.e.,

$$f[t, t+1, \dots, t+k] = \frac{(e-1)^k}{k!} e^t.$$

We need to prove it for $k+1$:

$$f[t, t+1, \dots, t+k, t+k+1] = \frac{(e-1)^{k+1}}{(k+1)!} e^t.$$

$$f[t, t+1, \dots, t+k, t+k+1] = \frac{f[t+1, t+2, \dots, t+k+1] - f[t, t+1, \dots, t+k]}{(t+k+1) - t}.$$

$$f[t+1, t+2, \dots, t+k+1] = \frac{(e-1)^k}{k!} e^{t+1},$$

$$f[t, t+1, \dots, t+k] = \frac{(e-1)^k}{k!} e^t.$$

Thus:

$$f[t, t+1, \dots, t+k, t+k+1] = \frac{\frac{(e-1)^k}{k!} e^{t+1} - \frac{(e-1)^k}{k!} e^t}{1} = \frac{(e-1)^k}{k!} (e^{t+1} - e^t) = \frac{(e-1)^k}{k!} e^t (e-1).$$

Simplifying:

$$f[t, t+1, \dots, t+k, t+k+1] = \frac{(e-1)^{k+1}}{(k+1)!} e^t.$$

$\square$

Question2

$$f[0, 1, \dots, n] = \frac{(e-1)^n}{n!} e^0 = \frac{(e-1)^n}{n!}.$$

Thus:

$$\frac{(e-1)^n}{n!} = \frac{1}{n!} f^{(n)}(\xi).$$

Since $f^{(n)}(x) = e^x$:

$$(e-1)^n = e^\xi.$$

Taking the natural logarithm:

$$\xi = \ln((e-1)^n) = n \ln(e-1).$$

So, ξ is located to the right of $n/2$.

IV

Question1

x	$f(x)$	$f[x, x+1]$	$f[x, x+1, x+2]$	$f[x, x+1, x+2, x+3]$
0	5			
1	3	$\frac{3-5}{1-0} = -2$		
3	5	$\frac{5-3}{3-1} = 1$	$\frac{1-(-2)}{3-0} = 1$	
4	12	$\frac{12-5}{4-3} = 7$	$\frac{7-1}{4-1} = 2$	$\frac{2-1}{4-0} = \frac{1}{4}$

[1]

$$p_3(f; x) = f(0) + f[0, 1](x-0) + f[0, 1, 3](x-0)(x-1) + f[0, 1, 3, 4](x-0)(x-1)(x-3)$$

Simplifying:

$$p_3(f; x) = 5 - \frac{9}{4}x + \frac{1}{4}x^3.$$

Question2

$$p_3'(f; x) = -\frac{9}{4} + \frac{3}{4}x^2.$$

so, $x = \sqrt{3}$, when the derivative of the polynomial is zero. □

V

Question1

We assume

$$p_5(f; x) = \sum_{i=0}^5 a_i x^i$$

, and we have:

$$p_5(0) = f(0), \quad p_5(1) = f(1), \quad p_5'(1) = f'(1), \quad p_5''(1) = f''(1), \quad p_5(2) = f(2), \quad p_5'(2) = f'(2)$$

So, $a_0 = 0$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \\ 0 & 2 & 6 & 12 & 20 \\ 2 & 4 & 8 & 16 & 32 \\ 1 & 4 & 12 & 32 & 80 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{pmatrix} = \begin{pmatrix} 1 \\ 7 \\ 42 \\ 128 \\ 448 \end{pmatrix} \quad (1)$$

So, we have: $f[0, 1, 1, 1, 2, 2] = a_5 = 30$

Question2

$$f[0, 1, 1, 1, 2, 2] = a_5 = 30 = \frac{f^{(5)}(\xi)}{5!} \text{ So, } \xi = \sqrt{\frac{10}{7}} \quad \square$$

VI

Question1

$$\begin{array}{c|cccc} 0 & 1 & & & \\ 1 & 2 & 1 & & \\ 1 & 2 & -1 & -2 & \\ 3 & 0 & -1 & 0 & \frac{2}{3} \\ 3 & 0 & 0 & \frac{1}{2} & \frac{-5}{36} \end{array}$$

$$p_4(f; x) = 1 + 1 * (x - 0) + -2 * (x - 0)(x - 1) + \frac{2}{3} * (x - 0)(x - 1)^2 + \frac{-5}{36} (x - 0)(x - 1)^2(x - 3)$$

$$p_4(f; 2) = 1 + 2 - 2 * 2 + \frac{2}{3} * 2 - \frac{5}{36} * 2 * (-1) = \frac{11}{18}$$

Question2

$$\exists \xi \in (0, 3), \text{ such that } |f(x) - p_4(f; x)| = \left| \frac{f^{(5)}(\xi)}{5!} (x - 0)(x - 1)^2(x - 3)^2 \right| \leq \frac{M}{120} |(x - 0)(x - 1)^2(x - 3)^2|$$

$$x = 2, R_4(2) \leq \frac{M}{60}$$

VII

forwards

$$\Delta f(x) = f(x + h) - f(x),$$

$$\Delta^{k+1} f(x) = \Delta \Delta^k f(x) = \Delta^k f(x + h) - \Delta^k f(x).$$

We use mathematical induction to prove the formula for the forward difference.

Base case: When $k = 1$,

$$\Delta f(x) = f(x + h) - f(x).$$

Inductive hypothesis: Assume the formula holds for some k , i.e.,

$$\Delta^k f(x) = k! h^k f[x_0, x_1, \dots, x_k].$$

$$\Delta^k f(x+h) = k!h^k f[x_1, x_2, \dots, x_{k+1}].$$

Inductive step: Prove the formula holds for $k+1$

$$\Delta^{k+1} f(x) = \Delta(\Delta^k f(x)).$$

$$\Delta^{k+1} f(x) = \Delta^k f(x+h) - \Delta^k f(x) = k!h^k f[x_1, x_2, \dots, x_{k+1}] - k!h^k f[x_0, x_1, \dots, x_k].$$

$$f[x_0, x_1, \dots, x_{k+1}] = \frac{f[x_1, x_2, \dots, x_{k+1}] - f[x_0, x_1, \dots, x_k]}{x_{k+1} - x_0}.$$

$$\Delta^{k+1} f(x) = k!h^k \cdot \frac{f[x_1, x_2, \dots, x_{k+1}] - f[x_0, x_1, \dots, x_k]}{h} = (k+1)!h^{k+1} f[x_0, x_1, \dots, x_{k+1}].$$

$$\Delta^{k+1} f(x) = (k+1)!h^k \cdot \frac{f[x_1, x_2, \dots, x_{k+1}] - f[x_0, x_1, \dots, x_k]}{(k+1)h} = (k+1)!h^{k+1} f[x_0, x_1, \dots, x_{k+1}].$$

□

backwards

Similarly, we can prove the formula for the backward difference.

VIII

$$\begin{aligned} f[x_0, x_0, x_1, \dots, x_n] &= \lim_{x \rightarrow x_0} f[x_0, x_0, x_1, \dots, x_n, x] \\ &= \lim_{x \rightarrow x_0} \frac{f[x_1, \dots, x_n, x] - f[x_0, x_1, \dots, x_n]}{x - x_0} \\ &= \frac{\partial}{\partial x_i} f[x_0, x_1, \dots, x_n] \end{aligned}$$

Similarly, we can prove the formula for the general case.

$$\frac{\partial}{\partial x_i} f[x_0, x_1, \dots, x_n] = f[x_0, \dots, x_i, x_i, \dots, x_n]$$

IX

Let $g(x) = \frac{f(x)}{a_0}$, so we have

$$\max_{(a,b)} g(x) \geq \frac{1}{2^{n-1}} \left(\frac{b-a}{2}\right)^n.$$

So, $\min \max f(x) = a_0 \frac{1}{2^{n-1}} \left(\frac{b-a}{2}\right)^n$.

X

Since $T_n(x)$ is Chebyshev polynomial, assuming $T_n(x) = \cosh(n * \operatorname{arccosh}(x))$, it has $n+1$ extremum points. As we all know, $\|T_n(x)\|_\infty = 1$. Let $Q(x) = \frac{T_n(x)}{T_n(a)} - p(x)$, we have

$$Q(a) = \frac{T_n(a)}{T_n(a)} - p(a) = 0.$$

Now, we assume that the proposition to be proved is not true.

$$\exists p \in \mathbb{P}, \text{ such that } \max_{x \in [-1,1]} |p(x)| < \frac{1}{T_n(a)}.$$

$$Q(x'_k) = \frac{T_n(x'_k)}{T_n(a)} - p(x'_k), k = 0, 1, \dots, n.$$

Assume that $Q(x)$ has $n+1$ extremum points where the signs of the two neighboring values are different, then $Q(x)$ has at least n zeros. Moreover, we have $x = 1$ as a zero point. So, we have at least $n+1$ zeros. However, The degree of $Q(x)$ is n , which means it has n zeros most. Therefore, $Q(x) \equiv 0, \max_{[-1,1]} p(x) = \frac{1}{T_n(x)}$, which contradicts the assumption. $\square$

XI

Proof:

$$b_{n,k}(t) = \binom{n}{k} t^k (1-t)^{n-k}.$$

We need to prove:

$$b_{n-1,k}(t) = \frac{n-k}{n} b_{n,k}(t) + \frac{k+1}{n} b_{n,k+1}(t).$$

First, we have:

$$b_{n-1,k}(t) = \binom{n-1}{k} t^k (1-t)^{n-1-k}.$$

$$b_{n,k}(t) = \binom{n}{k} t^k (1-t)^{n-k}.$$

$$b_{n,k+1}(t) = \binom{n}{k+1} t^{k+1} (1-t)^{n-k-1}.$$

Now we need to prove:

$$\binom{n-1}{k} = \frac{n-k}{n} \binom{n}{k} (1-t) + \frac{k+1}{n} \binom{n}{k+1} t$$

$$\begin{aligned} \binom{n-1}{k} &= \frac{n-k}{n} \binom{n}{k} (1-t) + \frac{k+1}{n} \binom{n}{k+1} t \\ &= \frac{n-k}{n} \frac{n!}{k!(n-k)!} (1-t) + \frac{k+1}{n} \frac{n!}{(k+1)!(n-k-1)!} t \\ &= \frac{(n-k)(k+1)n! + (k+1)n!(n-k)}{n(k+1)!(n-k)!} \\ &= \frac{(n-1)!}{k!(n-k-1)!} \end{aligned}$$

The proof is completed.

XII

Proof:

$$\begin{aligned}\int_0^1 \binom{n}{k} t^k (1-t)^{n-k} dt &= \binom{n}{k} \int_0^1 t^k (1-t)^{n-k} dt \\ &= \binom{n}{k} \left(\left[-t^k \cdot \frac{(1-t)^{n-k+1}}{n-k+1} \right]_0^1 - \int_0^1 k t^{k-1} \cdot \frac{(1-t)^{n-k+1}}{n-k+1} dt \right) \\ &= 0 + \frac{k}{n-k+1} \int_0^1 \binom{n}{k} t^{k-1} (1-t)^{n-k+1} dt\end{aligned}$$

We repeatedly use the distribution integration formula k times, and it remains:

$$\int_0^1 \binom{n}{k} t^k (1-t)^{n-k} dt = \frac{k!(n-k)!}{(n)!} \int_0^1 t^0 (1-t)^n dt \int_0^1 t^0 (1-t)^n dt = \int_0^1 (1-t)^n dt = \frac{1}{n+1}$$

So, we have:

$$\int_0^1 b_{n,k}(t) dt = \binom{n}{k} \cdot \frac{k!(n-k)!}{(n+1)!} = \frac{1}{n+1}.$$

The proof is completed.

Acknowledgement

References

- [1] DeepSeek AI. *DeepSeek*. <https://www.deepseek.com/>. Accessed: 2024-10-03.